

Department Overview

- **Department Name:** Department of Environmental Science
- **Faculty:** Faculty of Life Sciences & Informatics
- **History:** The undergraduate BS program was initiated in 2014. The department has since expanded to offer a postgraduate MS Environmental Science program.
- **Core Focus:** Training students to identify principle-based solutions to society's emerging environmental issues with a focus on sustainable development. The curriculum is mapped according to national and international standards while addressing regional market demands.
- **Facilities:** Features a state-of-the-art research laboratory explicitly built to support postgraduate and research students.
- **Career Options:** Graduates can explore professional pathways in a variety of local, national, and international sectors including:
 - Academic Institutions
 - Local Government Bodies
 - Environmental Protection Agency (EPA)
 - Water and Power Development Authority (WAPDA)
 - Water and Sanitation Authority (WASA)
 - Department of Planning and Development
 - Department of Health
 - Oil and Gas Development Authority & Heavy Processing Industries
 - International multidisciplinary research setups via postgraduate scholarship routes.

1. Bachelor of Science in Environmental Science (BS Environmental Science)

Admission Requirements

- **Academic Background:** Intermediate (Pre-Medical / Pre-Engineering) or an equivalent qualification from any recognized board passed with at least **45% marks**, OR a Diploma of Associate Engineer (DAE) in a relevant discipline with a minimum of **45% marks**.

Degree Requirements

- **Total Credit Hours:** 124 – 136
- **Total Courses Required:** 40 – 44

- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour structures denote Total Credits (Theory - Lab).

- **1st Semester:** Introduction to Environmental Science 3(3-0), Fundamental of Biology-I 3(2-1), Basic Chemistry 3(2-1), Mathematics-1 3(3-0), Reading and Writing Skills 3(3-0), Pakistan Studies 2(2-0).
- **2nd Semester:** Introduction to Earth Science 3(2-1), Fundamentals of Biology-II 3(2-1), Introduction of Statistics 3(3-0), Sociology 3(3-0), Academic Reading and Writing 3(3-0), Islamic Studies / Ethics 2(2-0).
- **3rd Semester:** Introduction to Computer 3(2-1), Environmental Chemistry 3(2-1), Environmental Physics 3(2-1), Introduction to Economics 3(3-0), Communication and Presentation Skills 3(3-0), Introduction to Philosophy 3(3-0).
- **4th Semester:** Fundamentals of Ecology 3(3-0), Environmental Microbiology 3(2-1), Environmental Pollution 3(3-0), Climatology 3(3-0), Introduction of Psychology 3(3-0), Biodiversity and Conservation 3(3-0).
- **5th Semester:** Applied Ecology 3(2-1), Environmental Toxicology 3(2-1), Environmental Profile of Pakistan 3(3-0), Population and Environment 3(3-0), Analytical Techniques in Environmental Science 3(1-2), Elective-I 3(2-1).
- **6th Semester:** GIS and Remote Sensing 3(2-1), Environmental Economics 3(3-0), Environmental Monitoring 3(2-1), Environmental Management Systems 3(3-0), Elective-II 3(3-0).
- **7th Semester:** Climate Change 3(3-0), Environmental Impact Assessment 3(2-1), Research Methodology 3(3-0), Elective-III 3(3-0), Elective-IV 3(2-1).
- **8th Semester:** Environmental Governance 3(3-0), Health and Environment 3(3-0), Pollution Control Technologies 3(3-0), Final Year Project 6.

Available Undergraduate Electives Pool

1. Water Resources Management 3(2-1)
2. Soil and Environment 3(3-0)
3. Occupational Health and Safety 3(2-1)
4. Solid Waste Management 3(3-0)
5. Ecotourism 3(3-0)
6. Air and Noise Pollution 3(3-0)
7. Hydrology 3(3-0)
8. Disaster Risk Management 3(3-0)
9. Pollutant Movement in Soil 3(3-0)
10. Energy and Environment 3(3-0)
11. Agro-ecology 3(3-0)
12. Environmental Biotechnology 3(3-0)
13. Project Management 3(3-0)

- 14. Environmental Impact Analysis 3(3-0)
- 15. Urban Environmental Management 3(3-0)
- 16. Coastal Environmental Management 3(3-0)

2. Master of Science in Environmental Science (MS Environmental Science)

Admission Requirements

- **Academic Background:** An M.Sc. degree in Environmental Science or Biological Sciences with at least 50% marks under an annual grading system, OR a 4-year BS degree in Biological, Chemical, or Environmental Science.
- **Academic & Testing Minimums:** Minimum graduation CGPA $\geq$ 2.0 and a minimum threshold score of **50** on the GAT exam.

Degree Requirements

- **Total Credit Hours:** 30
- **Coursework Component:** 8 to 10 graduate-level courses.
- **Mandatory Dissertation:** Candidates must successfully complete and defend a minimum **6 credit hour research thesis** to qualify for the MS degree award. Minimum required academic status is a cumulative CGPA $\geq$ 2.5.